

Figure 2.3 The first step in creating a data visualization is getting the data.

importing and working with this data (the first step shown in figure 2.3). One core difference between these formats is how they model data. JSON and XML provide the capacity to encode nested relationships in a way that delimited formats like CSV don't. Another difference is that `d3.csv()` and `d3.json()` produce an array of JSON objects, whereas `d3.xml()` creates an XML document that needs to be accessed in a different manner.

FILE FORMATS

D3 has five functions for loading data that correspond to the five types of files you'll likely encounter: `d3.text()`, `d3.xml()`, `d3.json()`, `d3.csv()`, and `d3.html()`. We'll spend most of our time working with `d3.csv()` and `d3.json()`. You'll see `d3.html()` in the next chapter, where we'll use it to create complex DOM elements that are written as prototypes. You may find `d3.xml()` and `d3.text()` more useful depending on how you typically deal with data. You may be comfortable with XML rather than JSON, in which case you can rely on `d3.xml()` and format your data functions accordingly. If you prefer working with text strings, then you can use `d3.text()` to pull in the data and process it using another library or code.

Both `d3.csv()` and `d3.json()` use the same format when calling the function, by declaring the path to the file being loaded and defining the callback function:

```
d3.csv("cities.csv",function(error,data) {console.log(error,data)});
```

The error variable is optional, and if we only declare a single variable with the callback function, it will be the data:

```
d3.csv("cities.csv",function(d) {console.log(d)});
```

You first get access to the data in the callback function, and you may want to declare the data as a global variable so that you can use it elsewhere. To get started, you need a data file. For this chapter we'll be working with two data files: a CSV file that contains data about cities and a JSON file that contains data about tweets, as shown in the following listings.

Listing 2.1 File contents of cities.csv

```
"label","population","country","x","y"
"San Francisco", 750000,"USA",122,-37
"Fresno", 500000,"USA",119,-36
"Lahore",12500000,"Pakistan",74,31
"Karachi",13000000,"Pakistan",67,24
"Rome",2500000,"Italy",12,41
```